Badge Hacking for Beginners

Nathan Jones

November 2026



LightTurtles

Welcome!

- Make sure you have a way to connect to the badge. Either:
  - Install Thonny
  - Use the browser-based Viper IDE with a Chrome-based web browser (Chrome, Edge, Brave, Opera)
  - Install `mpremote` (if you prefer the command line)
- Clone the badge's firmware repo
- Get an I2C sensor and a cable (graciously donated by DigiKey !)
- Get to know the folks around you!

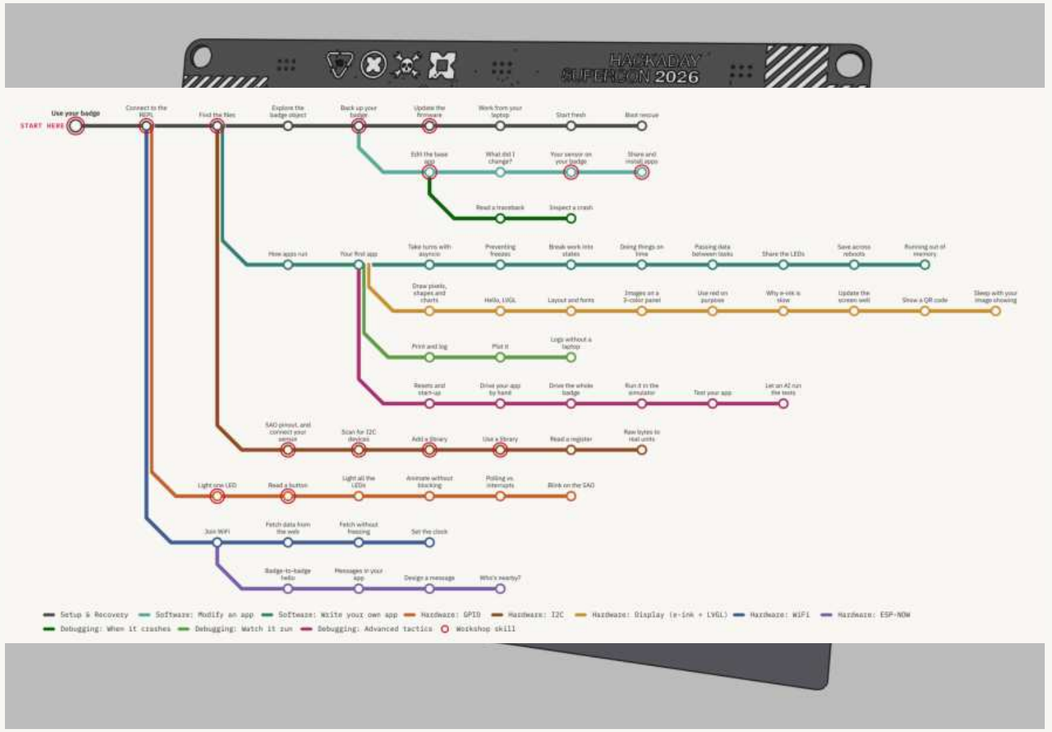
The Plan

1. Intro (20 min)
2. Skills map (90 min)
    1. 14 skills
    2. The result
3. Wrap-up (10 min)
    1. Plan your hack!
    2. Provide some feedback



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What was the first electronic project you built yourself?

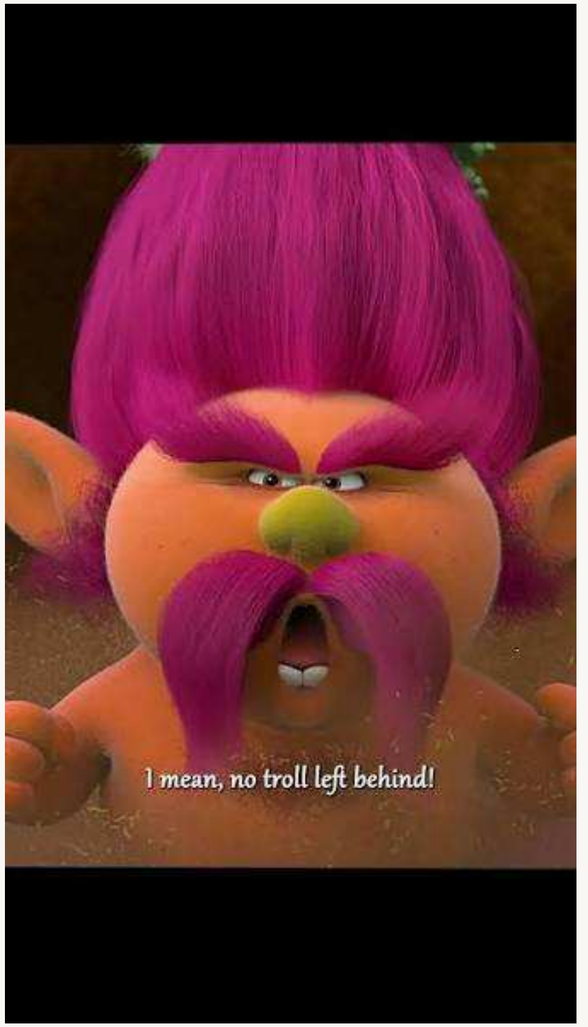


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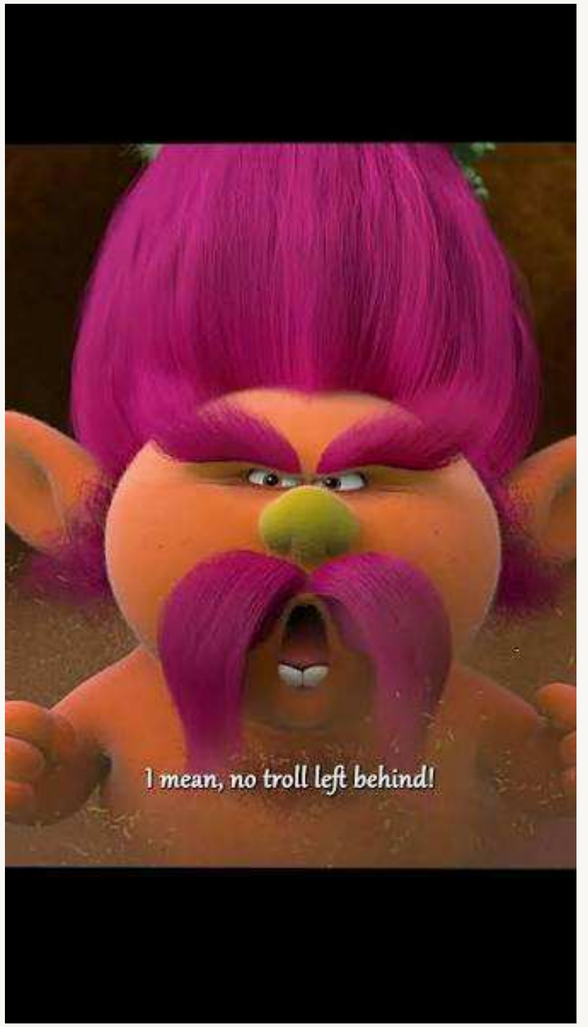
- Master's of Computer Engineering from NC STATE UNIVERSITY
- 15 years in the U.S.ARMY as a Military Intelligence Officer
  - 5 of those at West Point
- Selected works
  - "Enter the Matrix"
  - "Inside the Voja4"
  - "How Hardware Gets Hacked"
- Owner/operator LightTurtles, LLC — Embedded Systems Design & Education (lightturtles.github.io)

Three rules



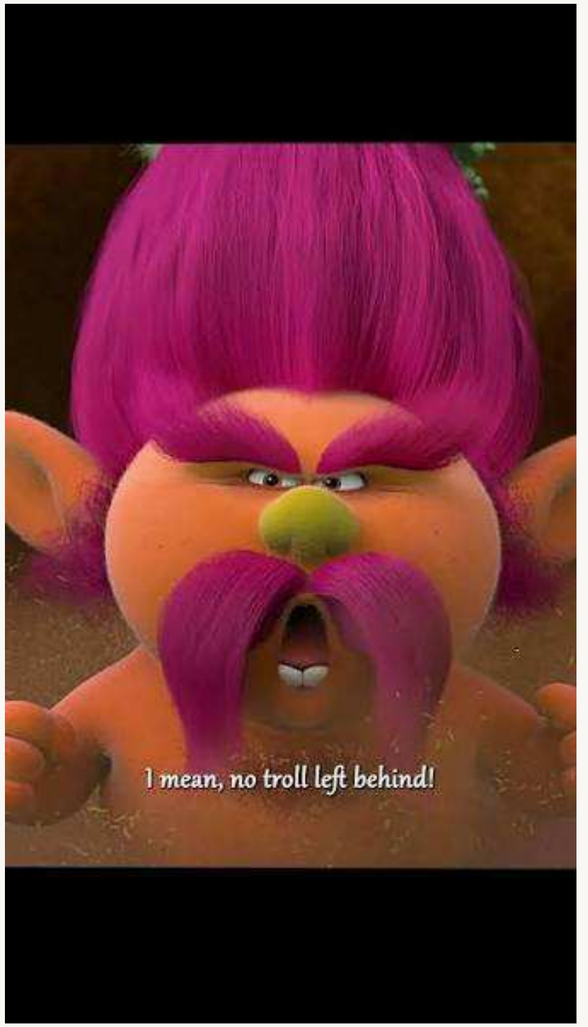
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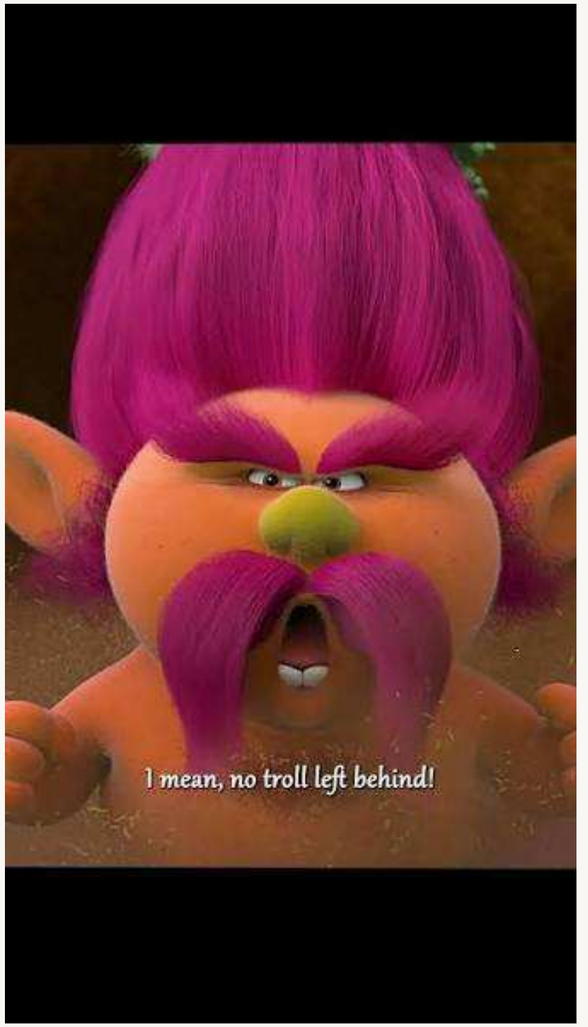
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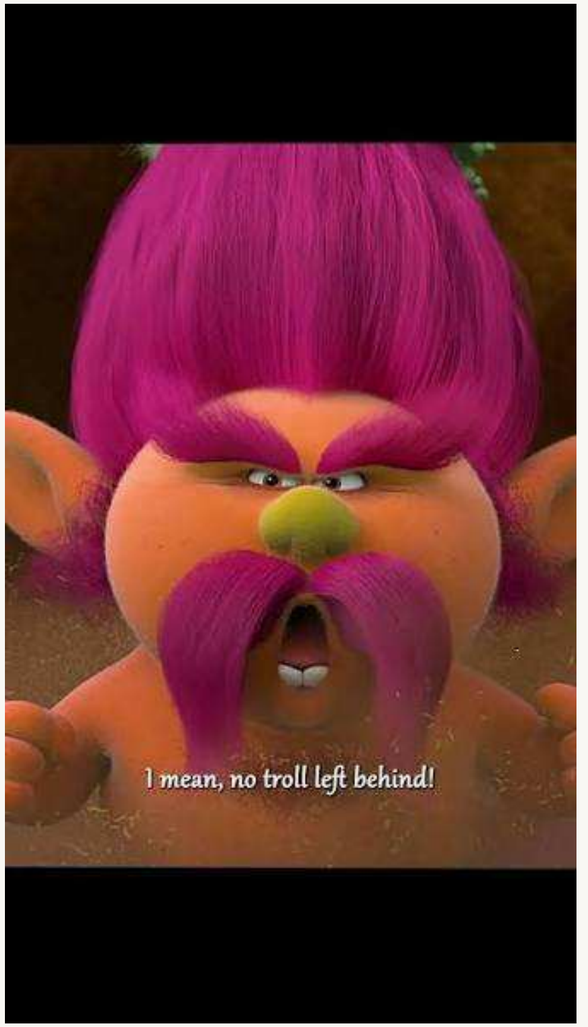
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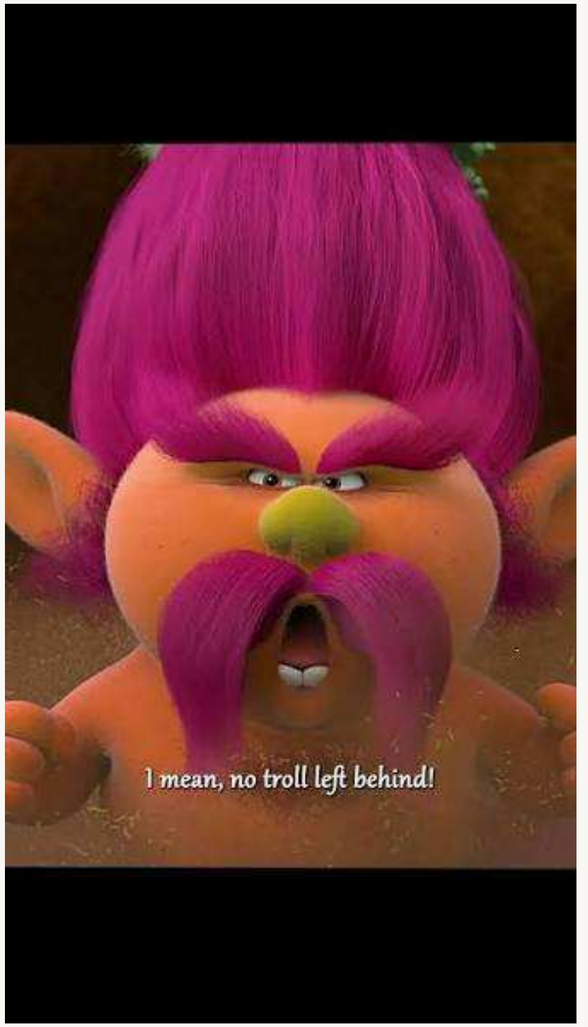
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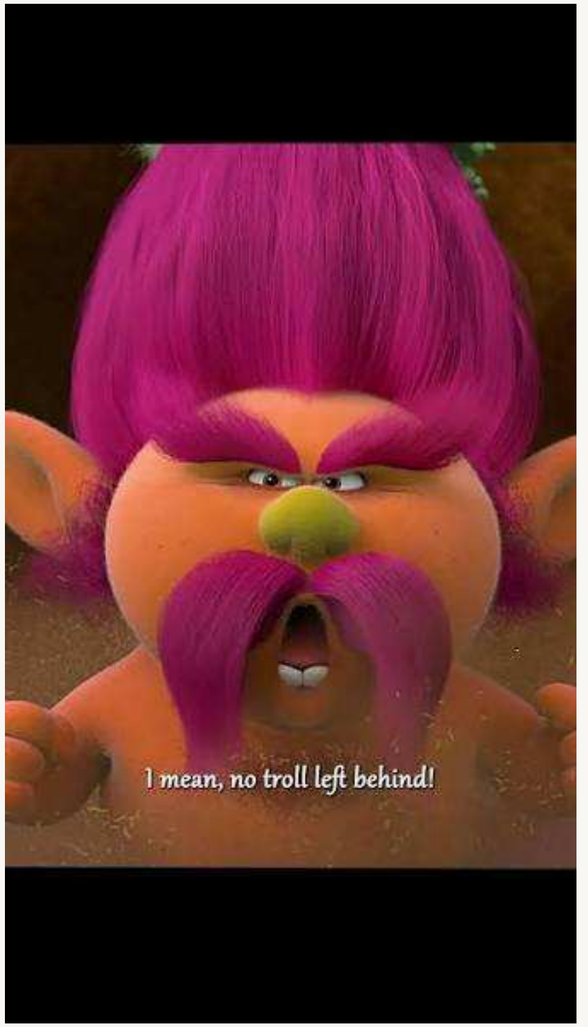
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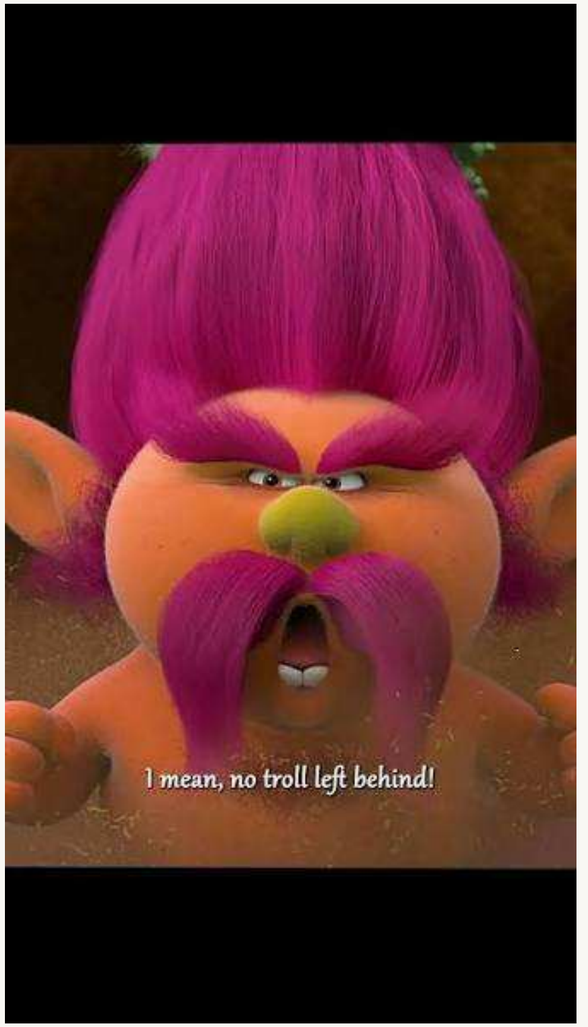
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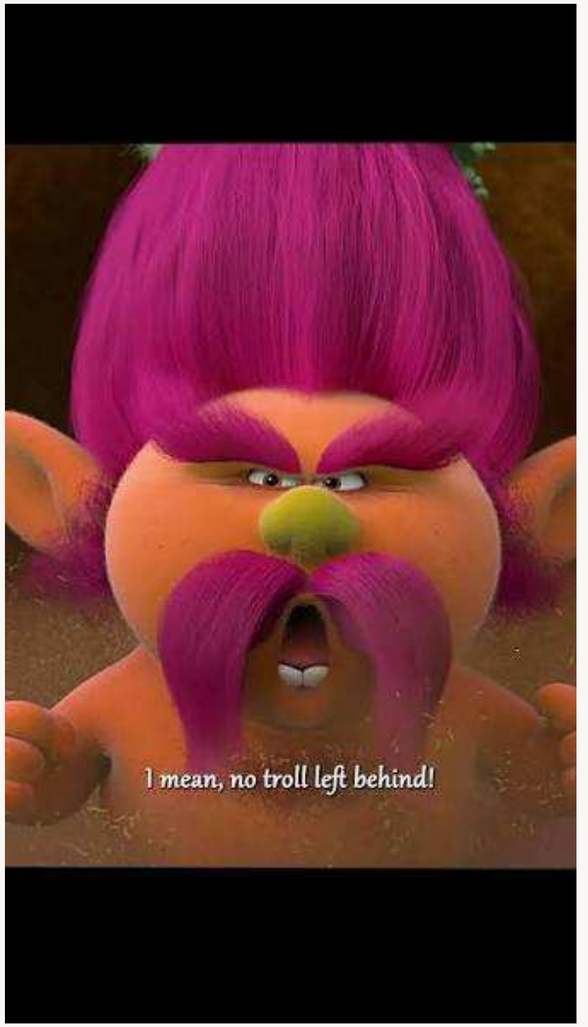
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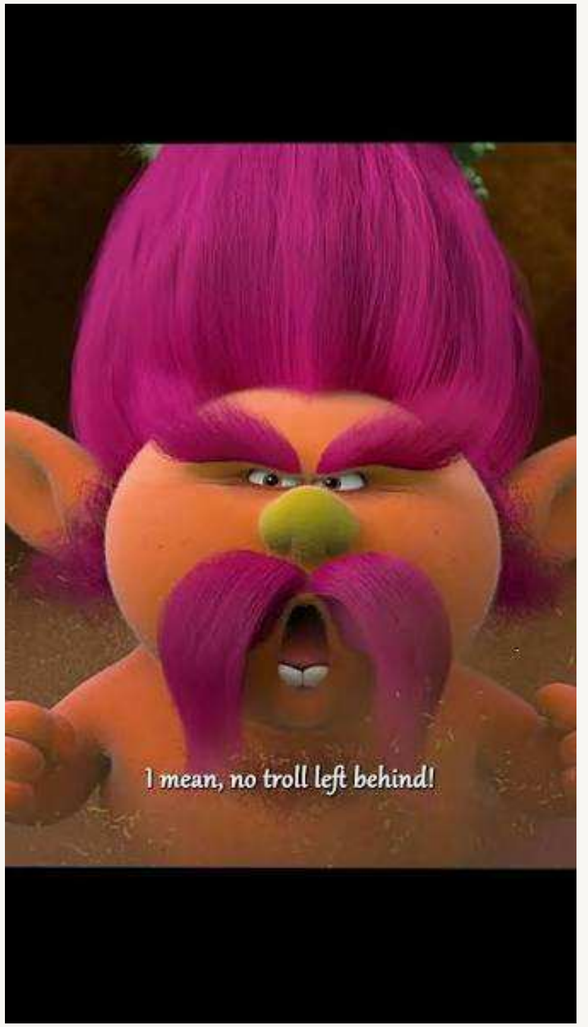
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  - So doing X makes Y happen.



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  - Wait, what if I…
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  - So doing X makes Y happen.
  - Here’s a graphic I drew about how this works.

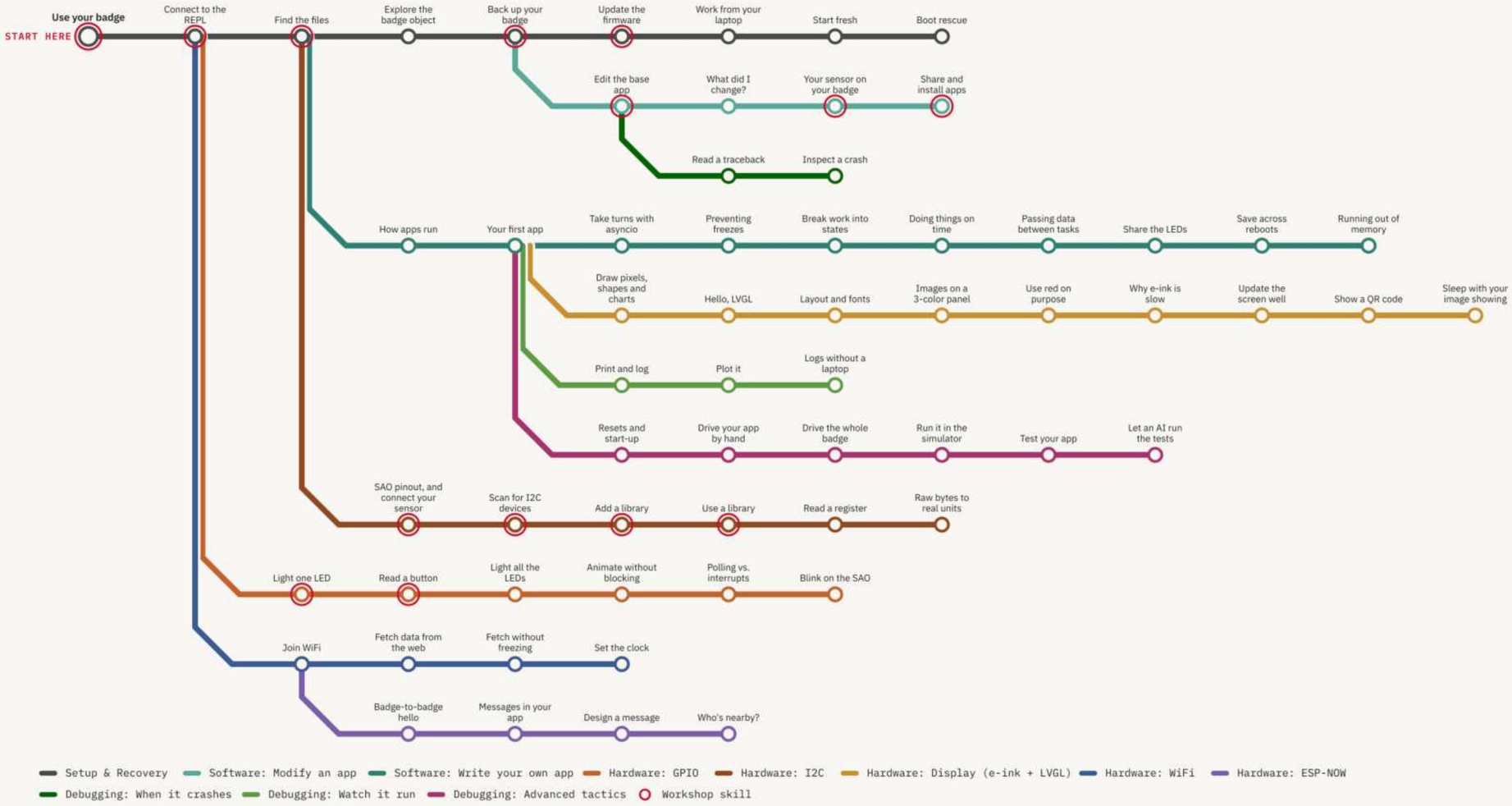


The 2026 Supercon Badge

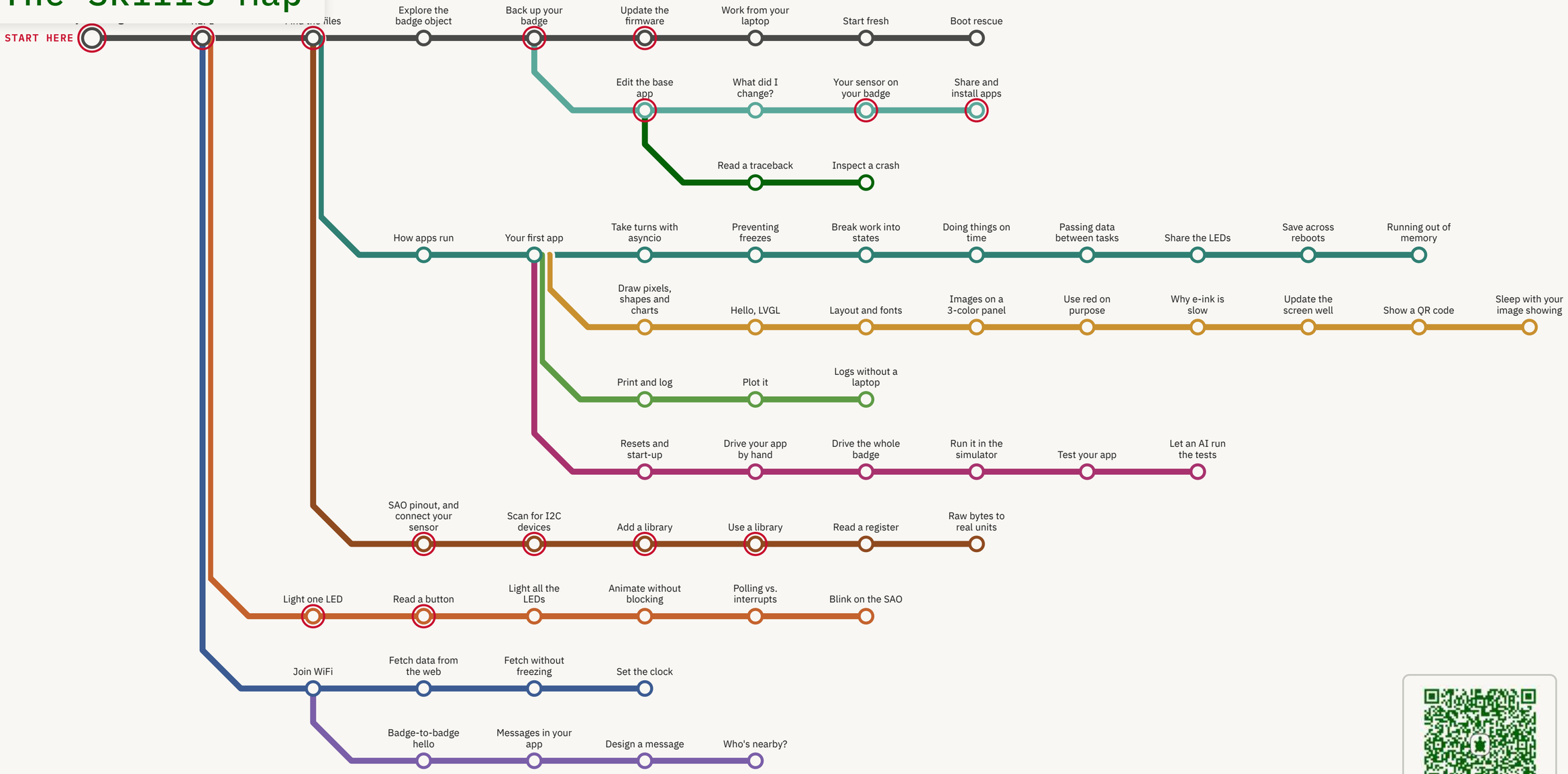


The Skills Map

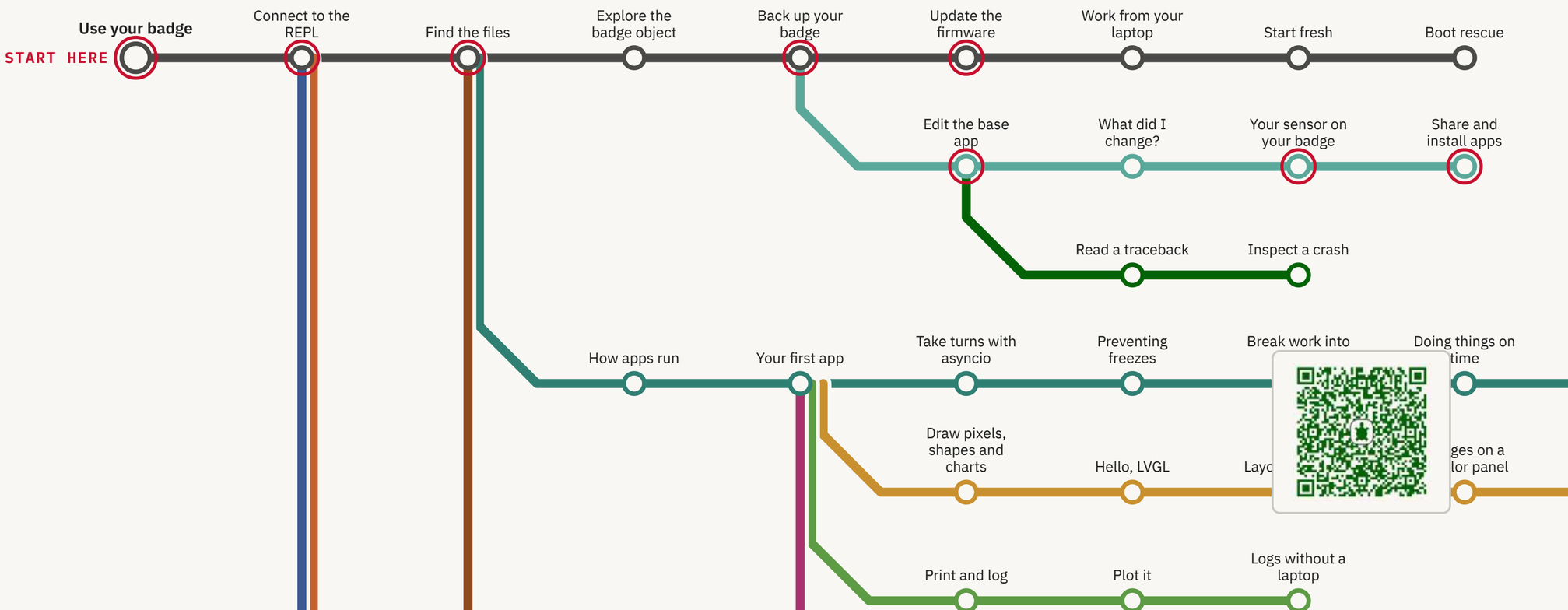
https://lightturtles.github.io/play/supercon26



The Skills Map

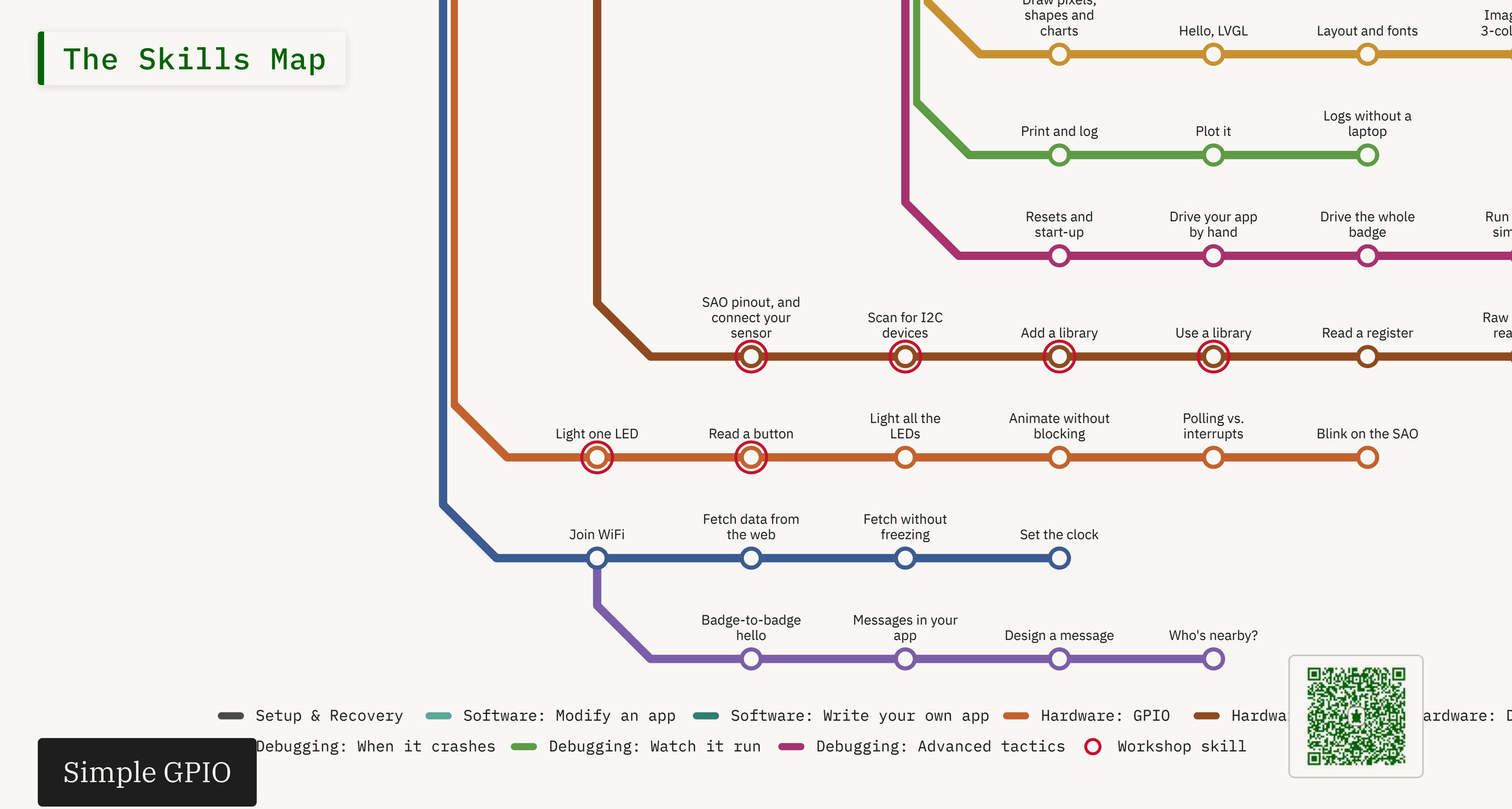


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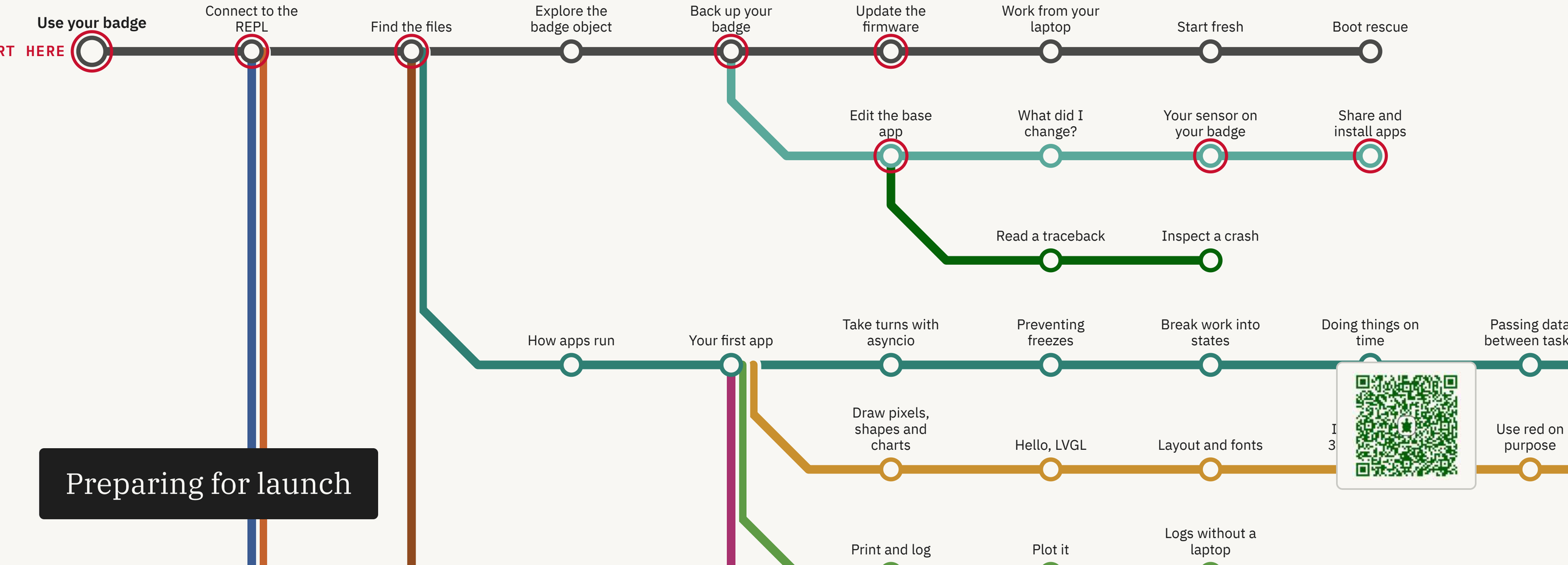
Badge orientation

The Skills Map

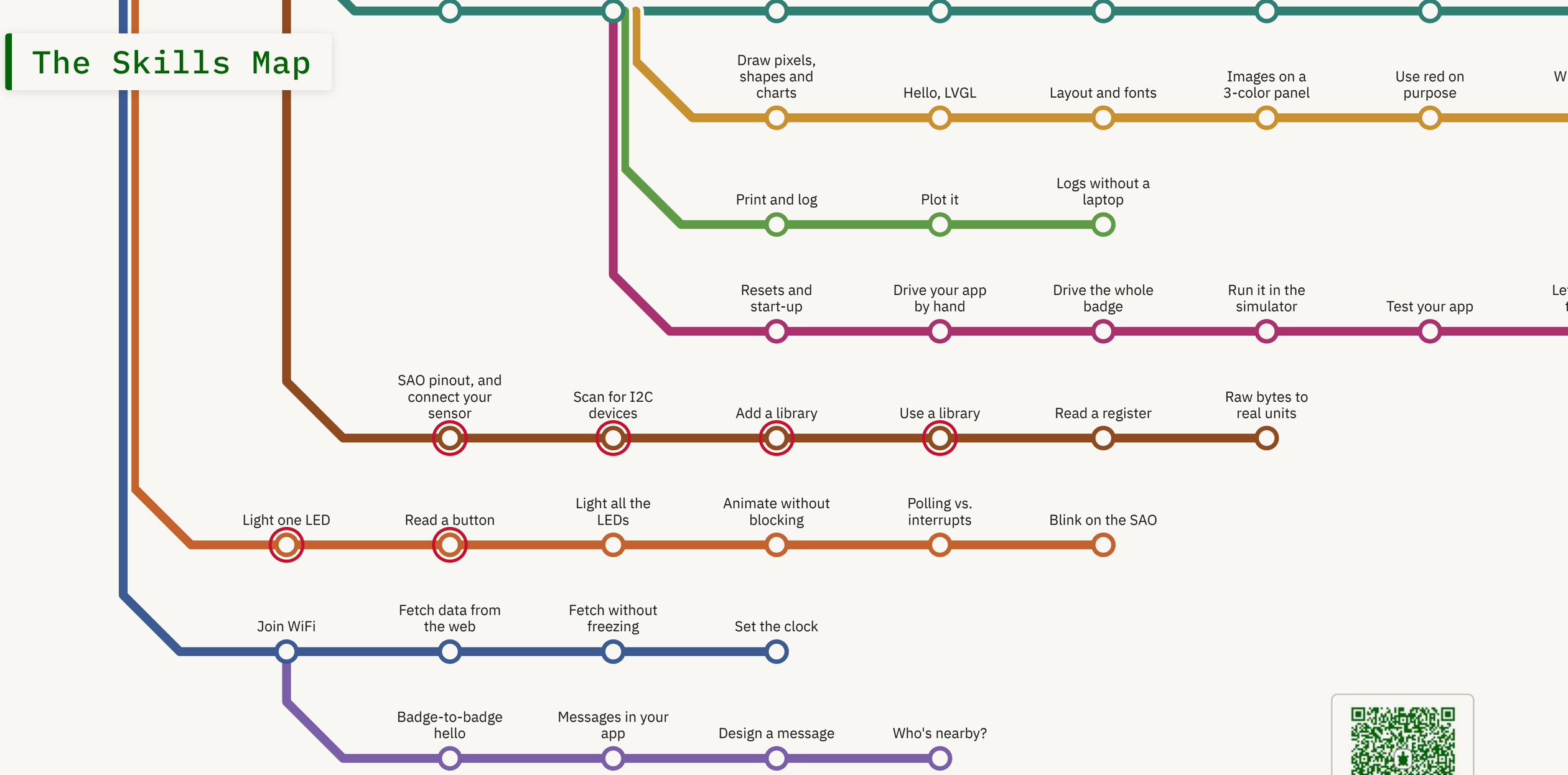


Simple GPIO

The Skills Map



The Skills Map

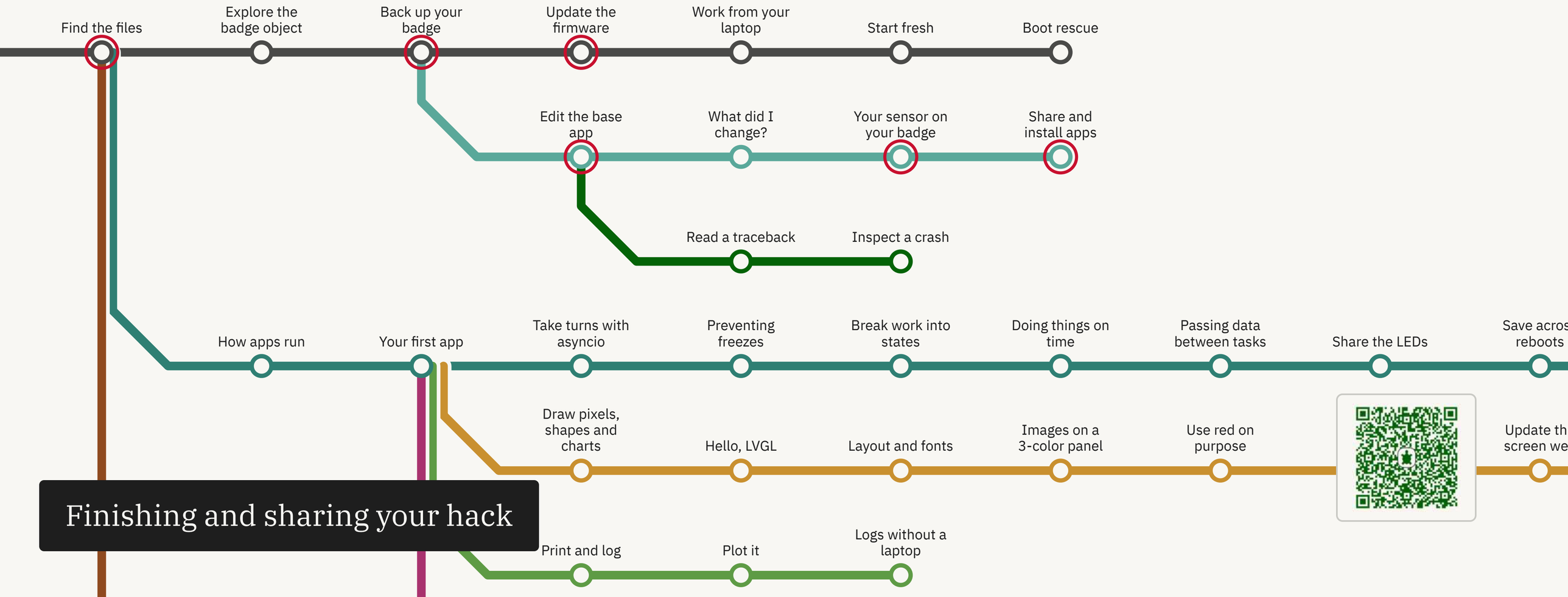


I2C sensors

: Modify an app · Software: Write your own app · Hardware: GPIO · Hardware: I2C · Hardware: Display · Hardw

Debugging: Watch it run · Debugging: Advanced tactics · Workshop skill

The Skills Map



Survey



